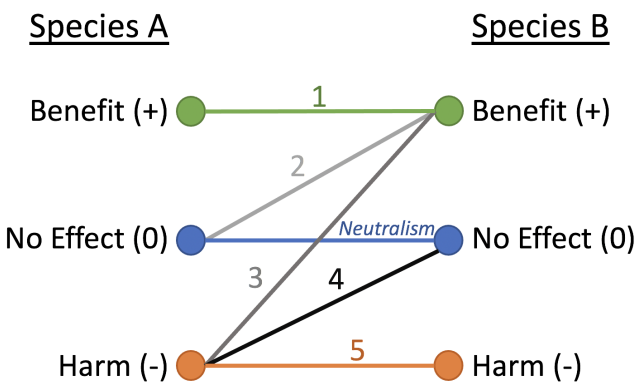
Community Ecology Study Guide

This study guide provides examples for the topics that you might see on the final exam over community ecology and species interactions. This is NOT an exhaustive list of all content that might be covered on the exam. Use your notes from lecture, the lecture slides and video on Canvas, and your homework questions to review all the topics that have been covered.

1. Fill in the blanks: Interactions with other species influence the [blank1] niche of a species, which relates to [blank 2] interactions.
   blank1 options: fundamental; realized
   blank2 options: biotic, abiotic

2. Which of the following would be considered symbiotic interactions? *Select all that apply*
   a. A legume, and the *Rhizobium* bacterial nodules in its roots
   b. A clown fish that lives in the tentacles of an anemone
   c. A tongue-eating isopod living in a fish's mouth
   d. A barnacle and the mussel species that outcompetes it for space and food
   e. A prairie dog and the burrowing owl that takes over its abandoned burrow
   f. A beaver and the fish that prefer the slow-moving water near the beaver dam

3. In the diagram below, each line represents a type of relationship between two species (A and B). The point of the line for each species shows how that species is affected by the interaction. Match the numbers on the diagram with the relationship they illustrate. One is already labeled for you: a relationship in which neither species is affected is called 'neutralism'.
   Terms: Mutualism, Exploitation, Symbiosis, Amensalism, Commensalism, Competition

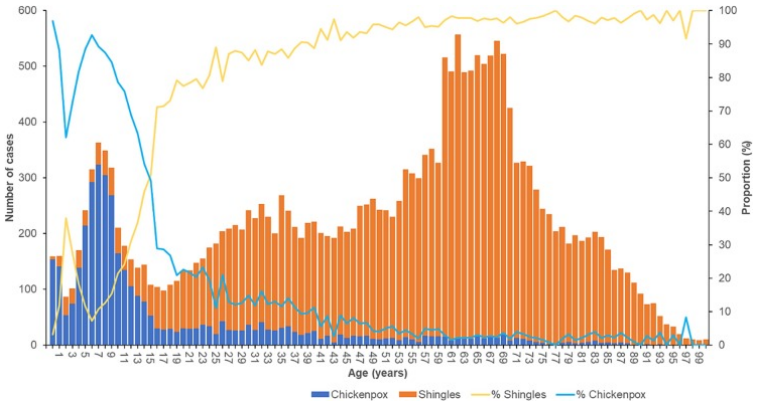


4. The Lotka-Volterra equations for defining predator and prey population sizes are included below. Consider an ecosystem in which lynx are the only predators of snowshoe hares and their interaction follows the Lotka-Volterra predictions. For each scenario, determine which term in the equations would be most directly affected by the scenario. *Note: a term may be used more than one, or not at all*

$$\frac{dV}{dt} = rV - pVP \qquad \frac{dP}{dt} = cpVP - d_pP$$

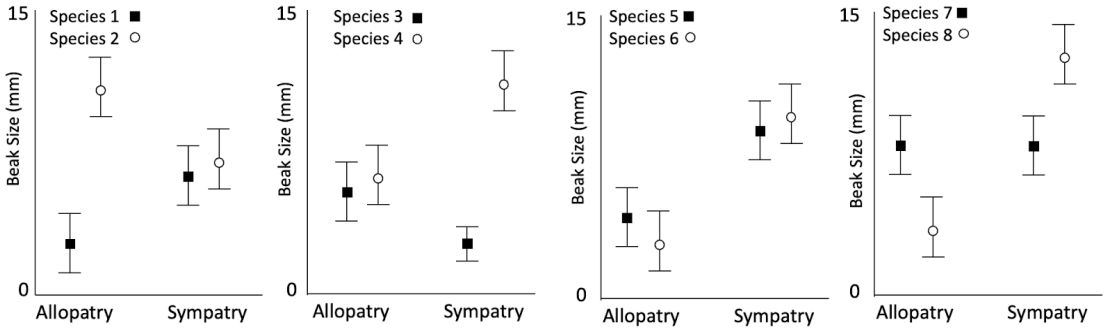
   a. A parasite is introduced to the ecosystem that significantly shortens the lifespan of lynx
   b. The number of hares that each lynx consumes in a year increases
   c. The number of offspring that a hare can produce in each litter increases
   d. The number of additional offspring that a lynx can support if it consumes more hares increases
   e. The carrying capacity of hares in the ecosystem increases

5. Match each scenario below with the competitive outcome it describes (competitive exclusion, resource partitioning, or character displacement).
    a. Wild turkey are found extensively throughout Yolo County, except in regions like Winters where the smaller, but more aggressive wild peafowl are found
    b. In tidal zones, shorebirds forage at different water depths, with plovers foraging in the exposed wet sand higher up in the shore, flamingoes foraging at deeper regions, and avocets foraging in between.
    c. When honeybees were introduced to North America, their larger size and more effective foraging abilities prevent many native bee species from foraging in regions where honeybee hives are found
6. The graph below shows the number (left y axis, blue and orange bars) and proportion (right y axis, teal and yellow lines) of Varicella Zoster Virus (VZV) cases by age (x axis) in Queensland, Australia from the years 2017-2021 (taken from Shrestha et al 2023). Blue bars represent the illness known as chickenpox (caused by initial contraction of VZV) and orange bars represent shingles (caused by a resurgence of the dormant virus in the body of individuals who previously contracted chickenpox). A vaccine for VZV was developed in the late 1990s, which shows 95% protection from contracting VZV as chickenpox in childhood and 100% prevention of serious cases. Prior to the vaccine, hospitalizations and deaths due to chickenpox were rare, but did occur (about 0.4% of cases resulted in hospitalization and 100-150 deaths per year just in the United States). The vaccine confers 80% protection of the resurgence of VZV as shingles in adults. Studies done by the World Health Organization indicate that herd immunity can be reached at 80% vaccination rates. The VZV vaccine is not required in Australia (nor in the United States at the federal level, though some states do require it, including California).



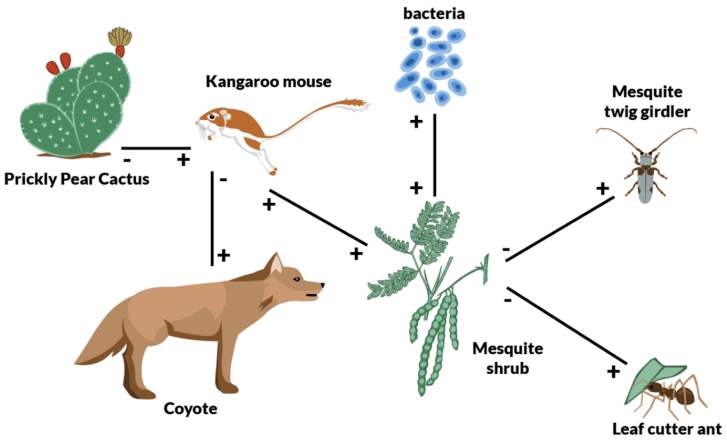
Given the scenario described above and the pattern of infections shown in the graph, what benefits would vaccination confer on a population? *Select all that apply*
    a. Decreased incidence of chickenpox in children
    b. Decreased incidence of shingles in adults
    c. Protection of unvaccinated individuals if vaccination rates exceed 80%
    d. Nearly complete elimination of severe cases of VZV
7. Both trait matching and evolutionary arms races are evolutionary processes in which the phenotypes (both morphological and behavioral) of two species evolve in response to one another. The difference between the two is that…
    a. trait matching occurs in interactions where one or both species benefit while evolutionary arms races occur in interactions where one or both species is harmed.
    b. trait matching occurs in herbivorous interactions while evolutionary arms races occur in predatory interactions.
    c. trait matching results in speciation while evolutionary arms races result in character displacement
    d. trait matching drives the evolution of homologous traits while evolutionary arms races drive the evolution of analogous traits

8. Each of the graphs below show beak shape (y axis) in two species (filled squares and open circles) when they occur sympatrically versus allopatrically (x axis). Each point represents the mean for that species in that habitat, and error bars show the variation in the population. Which of the species pairs shows evidence of character displacement having occurred in response to competition between the two species?



a. Species 1 and 2
b. Species 3 and 4
c. Species 5 and 6
d. Species 7 and 8
e. None of these species pairs show evidence of character displacement
f. All of these species pairs show evidence of character displacement

9. The interaction web below shows species interactions in a simplified desert shrub ecosystem. Each line connects two species and the plus and minus signs at each end indicate how each species is affected by the interaction. For example, Mesquite shrubs are legumes, which means they have a mutualistic interaction with soil bacteria in which the bacteria provide the plant with nitrogen and the plant provides the bacteria with sugar from photosynthesis. This interaction is represented by a line between the two species with a plus at each end indicating each species benefits from the interaction. There is not a line connecting the two insect species, the twig girdler and the leaf-cutter ant; however, we can determine that there is an interaction between these two species given that they rely on the same food source (Mesquite shrub).



<u>Fill in the blanks:</u>

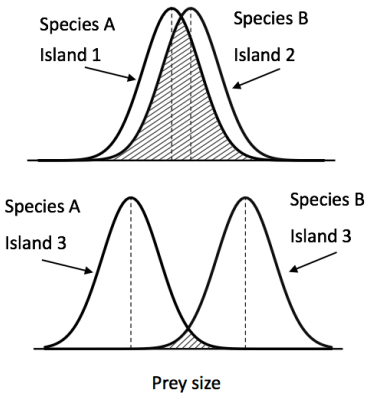
The interaction between these two species is [blank1] and would be represented by a line between the two insect species with a [blank2] sign for the effect on the leaf-cutter ant and a [answer3] sign for the effect on the twig girdler.

10. Define fundamental niche, realized niche, and carrying capacity.
    a. Draw a hypothetical fundamental niche for a species and use this diagram to depict how species interactions could shrink or expand the fundamental niche to produce the realized niche.

    b. Draw a hypothetical logistic growth curve for a species and use this diagram to depict how species interactions could increase or decrease the carrying capacity

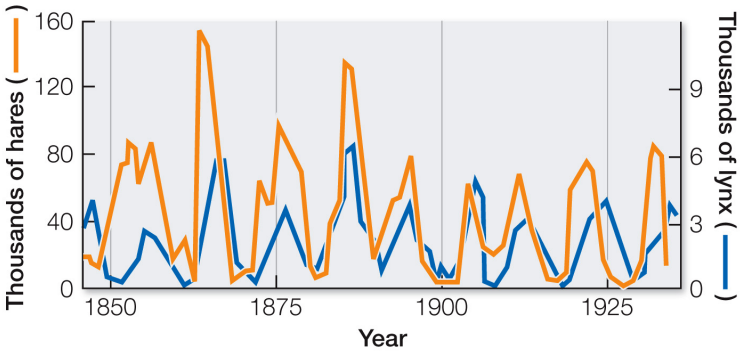
11. Compare and contrast the roles organisms can play in ecosystems: keystone species, foundation species, and ecosystem engineers.

12. Species A and B are lizards that eat insects. The figures at right show the preferred prey size of each species for populations living on Islands 1, 2, and 3. Determine if each of the statements below is true of these species on these islands.
    A. In allopatry, species A and B eat mostly the same prey
    B. In sympatry, resource partitioning has occurred
    C. In sympatry, species A and B diverge in prey choice
    D. Competition between species A and B affects prey choice in this group of islands
    E. Character displacement has decreased competition between species A and species B in allopatry



13. What is the competitive exclusion principle? How can organisms avoid the outcome predicted by the competitive exclusion principle?

14. The figure below shows the population size of hares in orange and lynx in blue in northern Canada from 1840-1940.
    a. What drives the oscillations in population size in these two species?
    b. Why do the high and low points of the lynx population generally fall shortly after the high and low points of the hare population?



LIFE: THE SCIENCE OF BIOLOGY 11e, Figure 55.9
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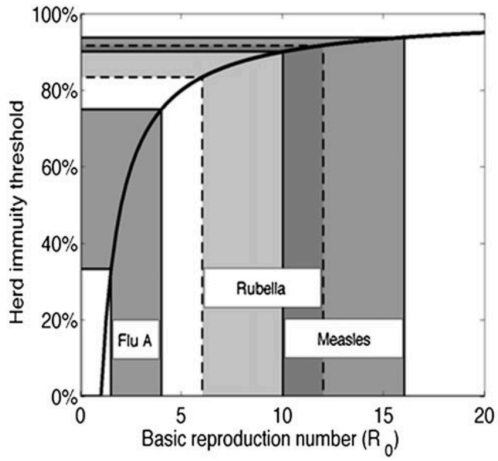
15. Write out the Lotka-Volterra equations for population growth in predators and prey.
    a. What term(s) in these equations represent reciprocal density dependence in population sizes of predators and prey?
    b. What term represents the growth rate of the predator population?
    c. Why are these equations formulated from the exponential growth equation and not from the logistic growth equation?

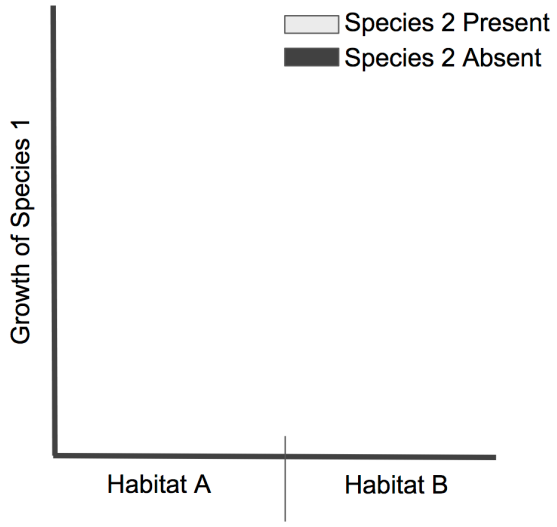
16. Why might natural selection favor a predator that is LESS efficient, or a disease/pathogen that is LESS virulent?

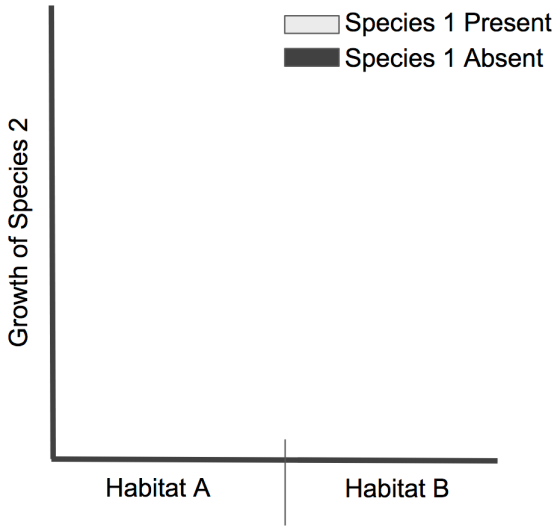
17. Using your understanding of density dependent interactions, explain how herd immunity protects susceptible individuals in the population from contracting a disease.

18. The graph Laci showed in lecture (and included below) shows how the R_0 value of different pathogens determines the degree of immunity that individuals in a population must meet in order to reach herd immunity. Use this graph to explain why vaccination for measles and rubella is required for public school children in the United States, but flu vaccination is not.



19. The graphs below show growth of Species 1 in Habitats A and B when Species 2 is present and absent (left graph) and the growth of Species 2 in Habitats A and B when Species 1 is present and absent (right graph). Complete the bar graphs with data that represent the following species interactions: Predation, Competition, Mutualism, Herbivory, Commensalism, and Parasitism





20. Describe how species interactions produce the following evolutionary trajectories in two species and explain how we know if these processes are occurring or have occurred in two species.
    a. Character displacement
    b. Trait matching
    c. Evolutionary 'arms race'